

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: *Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)*

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I(velarasan S/192571087) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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ANSWERS

The five debugging & optimisation scenarios below are addressed in order, each covering problem identification, the debugging approach, optimisation recommendations, and supporting analysis, in line with the rubric criteria.

Question 1: Enterprise WLAN & WWAN Connectivity

Scenario Summary

A six-storey smart office with 1,200+ users on Wi-Fi and IoT devices, plus field engineers on 4G/5G WWAN, is suffering slow speeds at peak hours, floor-to-floor disconnections, weak signal in conference rooms, video-conferencing drops, uneven AP loading, WWAN dead zones on the road, and devices still using outdated wireless security.

1. Problem Identification – Root Causes

- Insufficient access point (AP) density and poor placement, leaving conference rooms with weak signal and dead spots between floors.
- No load balancing/band steering — some APs are overloaded (too many associated clients) while neighbouring APs are underused.
- Overlapping or misconfigured Wi-Fi channels causing co-channel interference, especially in a dense multi-floor deployment.
- Lack of fast roaming (no 802.11k/v/r) so devices hold on to a weak AP too long when moving between floors, causing drops during calls.
- No QoS/traffic prioritisation for real-time video conferencing and VoIP traffic, so it competes equally with file downloads and backups.
- WWAN coverage gaps in rural/highway areas with no fallback (e.g., dual-SIM, multi-carrier failover, or store-and-forward buffering).
- Some endpoints still using outdated security protocols (WEP/WPA), which is both a security risk and can force lower PHY data rates on the whole cell.

2. Debugging Approach

1. Conduct a Wi-Fi site survey (heat map) per floor to locate coverage gaps, especially in conference rooms and stairwells.
2. Pull AP association/utilisation logs from the WLAN controller to identify overloaded vs. underutilised access points.
3. Run a channel scan to detect overlapping/interfering channels across adjacent APs and floors.
4. Capture and inspect video-conferencing sessions during drops to check for high jitter/latency correlating with AP handover events.
5. Audit connected devices for the security protocol/encryption in use (WEP/WPA/WPA2/WPA3).
6. Review WWAN signal logs and drive-test the routes field engineers travel to map dead zones.

3. Optimization Recommendations

- Increase AP density and reposition APs based on the heat map, giving dedicated coverage to conference rooms.
- Enable band steering and client load balancing so devices are pushed to 5 GHz/less-congested APs automatically.
- Re-plan the channel layout (non-overlapping 1/6/11 on 2.4 GHz, wider non-overlapping channels on 5 GHz) to cut interference.

- Enable fast/seamless roaming (802.11k/v/r) so calls do not drop while moving between floors.
- Apply WMM/QoS policies that prioritise VoIP and video-conferencing traffic ahead of bulk file transfers.
- Upgrade all devices to WPA3-Enterprise and disable legacy WEP/WPA associations.
- Deploy multi-carrier/dual-SIM WWAN routers with automatic failover, and enable store-and-forward on field apps so reports queue locally until connectivity returns.

4. Analysis / Justification

The connectivity symptoms map cleanly onto capacity and RF-planning faults rather than a single hardware failure: uneven AP load explains the peak-hour slowdowns, poor roaming explains floor-to-floor drops, and weak security explains the audit findings. Addressing AP density, channel planning, QoS, and WWAN redundancy together — rather than any one fix in isolation — restores both performance and security.

Question 2: ATM Network Performance in a Financial Enterprise

Scenario Summary

An ATM backbone carrying online banking, VoIP, video conferencing and real-time transactions is showing transaction delays, voice jitter/packet loss, and poor video quality. Some switches run at high utilisation, some Virtual Circuits (VCs) are congested while others are idle, and some applications use the wrong ATM service category.

1. Problem Identification – Root Causes

- Traffic-to-service-category mismatch: latency-sensitive voice/video traffic is likely running over UBR/ABR instead of CBR/rt-VBR, so it gets no delay/jitter guarantee.
- Uneven VC provisioning — bandwidth is statically assigned to VCs without traffic engineering, so some VCs are congested and others sit idle.
- No (or misconfigured) Connection Admission Control (CAC), allowing more traffic onto VCs than their negotiated Peak/Sustainable Cell Rate (PCR/SCR) can support.
- High switch utilisation at peak hours, causing cell queuing delay and cell loss at congested switches.

2. Debugging Approach

1. Check the ATM service category (CBR, rt-VBR, nrt-VBR, ABR, UBR) assigned to each application/VC against its actual traffic profile.
2. Pull per-VC utilisation and cell-loss/cell-delay-variation (CDV) statistics from switch management tools during peak hours.
3. Verify CAC and traffic contract parameters (PCR, SCR, MBS) configured versus the real offered load.
4. Trace congested switches/links to confirm whether the bottleneck is a specific VC, trunk, or switch backplane.

3. Optimization Recommendations

- Reassign traffic to the correct ATM service categories: CBR for voice, rt-VBR for interactive video, nrt-VBR for transactional banking data, UBR/ABR for background transfers.
- Re-provision VC bandwidth based on measured demand (traffic engineering) instead of static assignment, shifting capacity from idle VCs to congested ones.
- Enforce CAC so new connections are only admitted when sufficient bandwidth/QoS headroom exists, preventing over-subscription.
- Add/upgrade trunks or switch capacity at consistently high-utilisation points identified during debugging.
- Implement traffic shaping/policing at the network edge to keep each VC within its negotiated contract and reduce burst-induced congestion.

4. Analysis / Justification

Fixed-size 53-byte cells make ATM well-suited to guaranteed QoS, but only if traffic is placed in the category matching its sensitivity to delay and loss. The reported symptoms — jitter on voice, poor video, and slow transactions — are consistent with real-time traffic sharing best-effort categories and with static VC bandwidth that no longer matches actual demand, so correcting classification and re-engineering VC bandwidth directly targets the root cause rather than just adding raw bandwidth.

Question 3: MPLS Network Faults in a Global E-Commerce Backbone

Scenario Summary

An MPLS backbone linking HQ, regional offices, warehouses and cloud data centers is showing slow cloud access, delayed order processing, poor voice quality, and video interruptions. Some Label Switched Paths (LSPs) are heavily congested while others are underused, some routers have incorrect label mappings, and traffic engineering (TE) has not been configured.

1. Problem Identification – Root Causes

- Missing or disabled MPLS Traffic Engineering (TE), so traffic follows the IGP shortest path only, overloading some LSPs while alternate paths sit idle.
- Incorrect label bindings/mappings on certain routers, causing packets to be forwarded along suboptimal or wrong LSPs and increasing latency.
- No differentiated QoS classes across the MPLS backbone, so VoIP/video is not prioritised over ERP/order-processing or bulk data traffic.
- No Fast Reroute or load-sharing configured across equal-cost LSPs, concentrating load on fewer paths.

2. Debugging Approach

1. Use MPLS ping/traceroute (LSP ping) on each LSP to verify the label-switched path matches the intended route and confirm label bindings at each hop.
2. Check LDP/RSVP-TE label tables on suspect routers for mismatched or stale label mappings.
3. Review per-LSP bandwidth utilisation to identify which paths are congested and which are underused.
4. Inspect QoS class-of-service (CoS) configuration on ingress routers to see how voice, video, and data are currently marked and queued.

3. Optimization Recommendations

- Enable MPLS-TE with RSVP-TE (or segment routing TE) so LSPs are placed based on available bandwidth, not just IGP shortest path, spreading load across the backbone.
- Correct the faulty label mappings identified during debugging and re-verify with LSP ping/traceroute.
- Configure CoS-based queuing (e.g., EF for VoIP, AF4x for video, AF for ERP/order data, BE for bulk transfers) end-to-end across the MPLS cloud.
- Enable load-sharing/ECMP across multiple LSPs between the same endpoints, and configure Fast Reroute for automatic failover on link/node failure.
- Set up ongoing LSP utilisation monitoring/alerts so congestion is caught before it affects customers.

4. Analysis / Justification

Congested paths existing alongside idle paths is a textbook signature of an MPLS network running without traffic engineering — the IGP is choosing the same 'shortest' path for everything. Combined with incorrect label mappings misdirecting some flows, this fully explains the latency and jitter reported. Enabling TE-based path placement and

correcting label errors addresses the structural cause, while CoS queuing protects real-time voice/video once the paths themselves are balanced.

Question 4: Data Transmission Errors in an Automated Warehouse

Scenario Summary

Barcode scanners, RFID readers, and robots exchange data with a central server; the company is seeing incorrect shipment records, failed scans, and inventory mismatches. Increased transmission errors from electrical interference, signal attenuation, and congestion occur at peak hours, and some corrupted frames go undetected.

1. Problem Identification – Root Causes

- Weak error-detection scheme (e.g., simple parity or a weak checksum) that cannot reliably catch multi-bit burst errors typical of electrical/RF interference.
- Physical-layer issues — EMI from warehouse machinery and signal attenuation over distance — increasing the raw bit-error rate at the source.
- Network congestion at peak hours causing buffer overflows and dropped/corrupted frames in addition to line errors.
- No reliable retransmission (ARQ) scheme in place for frames that are detected as corrupted, so bad data slips through to the database.

2. Debugging Approach

1. Sample frames at the affected terminals and check the current error-detection method (parity bit / simple checksum / CRC) against known error patterns.
2. Measure bit-error rate and correlate spikes with specific times, machines, or cable runs to confirm EMI/attenuation as a source.
3. Check switch/AP buffer occupancy and drop counters during peak hours to quantify the congestion contribution.
4. Verify whether a retransmission (ARQ) mechanism exists and, if so, whether it is actually triggered when errors are detected.

3. Optimization Recommendations

- Replace weak error detection with CRC (e.g., CRC-32), which reliably detects burst errors far better than parity or simple checksums.
- Implement an ARQ scheme (Go-Back-N or Selective-Repeat) so any frame that fails the CRC check is automatically retransmitted rather than silently accepted.
- Improve the physical layer: use shielded/fiber cabling near heavy machinery, relocate or shield APs, and reduce cable run lengths to cut attenuation and EMI pickup.
- Segment the warehouse network (VLANs/additional switches or APs) to reduce congestion-driven buffer overflow during peak hours.
- Add application-layer checksums/hashes on shipment and inventory records as a final integrity check before database commit.

4. Analysis / Justification

Undetected corrupted frames causing wrong inventory data is a strong indicator that the existing detection scheme (likely parity/simple checksum) is too weak for the burst-error environment created by warehouse EMI and attenuation. Layering a strong CRC with an ARQ retransmission scheme addresses detection and recovery together, while physical-layer and congestion fixes reduce how often errors occur in the first place — together restoring end-to-end data accuracy.

Question 5: Flow Control Failure on an Online Examination Platform

Scenario Summary

A 10 Gbps central server sends question papers, images, video and acknowledgements to 15,000+ students on much slower broadband/mobile links. Peak-hour symptoms include incomplete downloads, duplicate packets, repeated retransmissions, receiver buffer overflow/application crashes, and poor throughput — pointing to ineffective flow control between the fast sender and slow, heterogeneous receivers.

1. Problem Identification – Root Causes

- Sender transmission rate is not adapted to each receiver's much slower link speed, so the server effectively floods low-bandwidth clients.
- Fixed or oversized transmission window that does not shrink for slow receivers, causing more data to be sent than the receiver's buffer can absorb (buffer overflow).
- Aggressive retransmission timers firing before slow acknowledgements arrive, producing duplicate packets and unnecessary retransmissions.
- No differentiation between fast and slow clients — one transmission strategy is applied to a highly heterogeneous population of 15,000 students.

2. Debugging Approach

1. Check whether the current mechanism uses a fixed window or a properly adaptive sliding window with dynamic receiver-advertised window size.
2. Capture traffic to a sample of slow clients to confirm receiver buffer overflow and measure actual RTT/loss versus the timeout values in use.
3. Review retransmission timeout (RTO) configuration against observed RTT variability across broadband vs mobile clients.
4. Check whether TCP window scaling and congestion control are enabled and correctly negotiated on both server and client sides.

3. Optimization Recommendations

- Use a dynamic sliding-window flow control mechanism so the receiver's advertised window (not a fixed sender rate) governs how much unacknowledged data is in flight per client.
- Enable TCP window scaling and tune RTO calculation (e.g., using measured RTT/variance) to reduce premature retransmissions and duplicate packets.
- Apply per-client rate pacing/throttling on the server so fast and slow clients each receive data matched to their own link capacity.
- Offload static content (question papers, images, videos) to a CDN or edge caching layer close to students, reducing direct load on the origin server.
- Use Selective-Repeat ARQ instead of simple Go-Back-N so only the specific lost/corrupted segment is retransmitted, not the whole window.
- Consider multicast delivery for identical content (e.g., the same question paper) to cut redundant transmissions during exam start times.

4. Analysis / Justification

The mismatch between a 10 Gbps sender and much slower, heterogeneous receivers is the classic condition sliding-window flow control is designed to solve: each receiver, not the sender, should determine how much unacknowledged data it can accept. Buffer overflow, duplicate packets, and excess retransmissions all disappear once the window size and pacing adapt per-connection to the receiver's real capacity, supplemented by CDN offload to reduce the absolute load the origin server must serve at exam start.